Development of the gonads

The Indifferent Embryo

- The gender of the embryo is determined genetically at the time of fertilization.

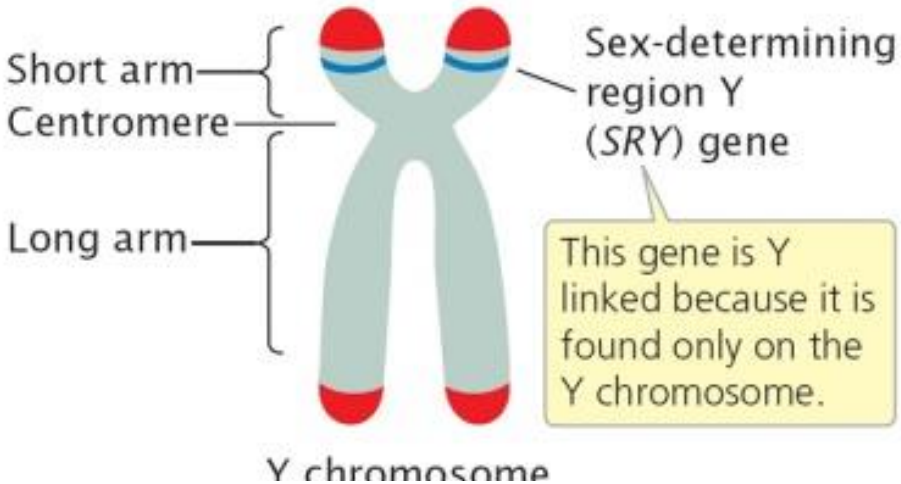
- However, the gonads do not acquire male or female morphological characteristics until the **seventh week** of development.





Development of the gonads

The Indifferent Embryo

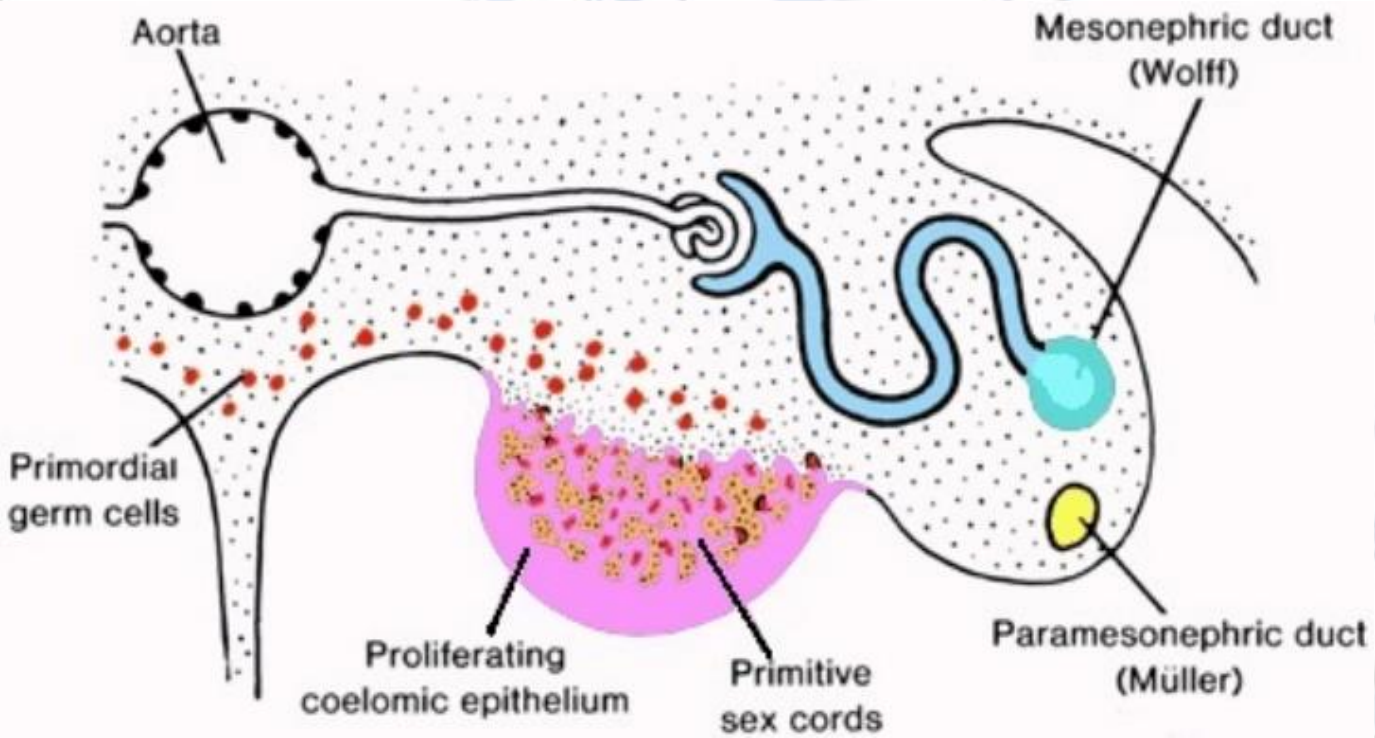


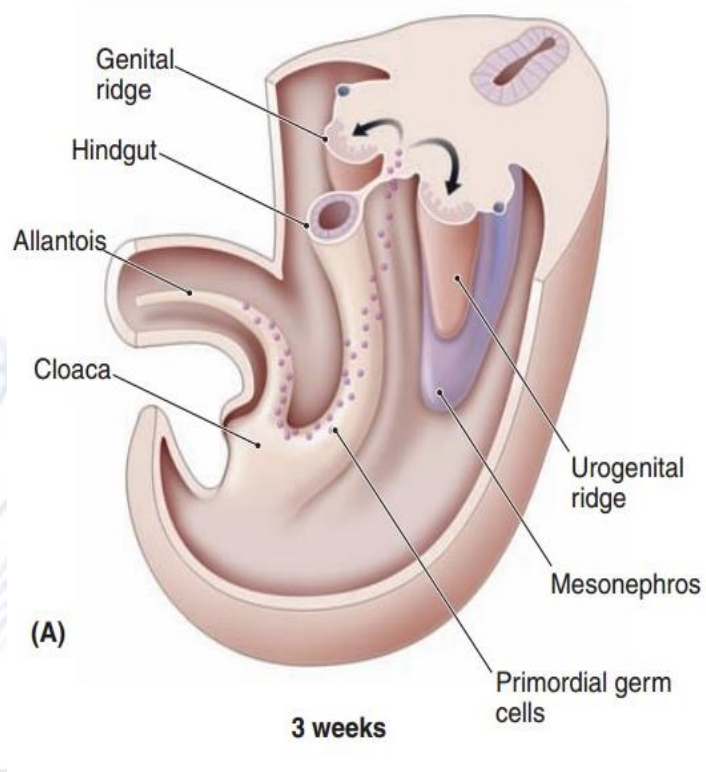
- Gender determination is a complex process that involves many genes.
- The key to phenotypic sexual differentiation is the Y chromosome, which contains the **SRY (sex-determining region on Y) gene** which is testis-determining gene. This gene produces **SRY protein** (testis determining factor).
- If it is present → male.
- In its absence → female.

Development of the gonads

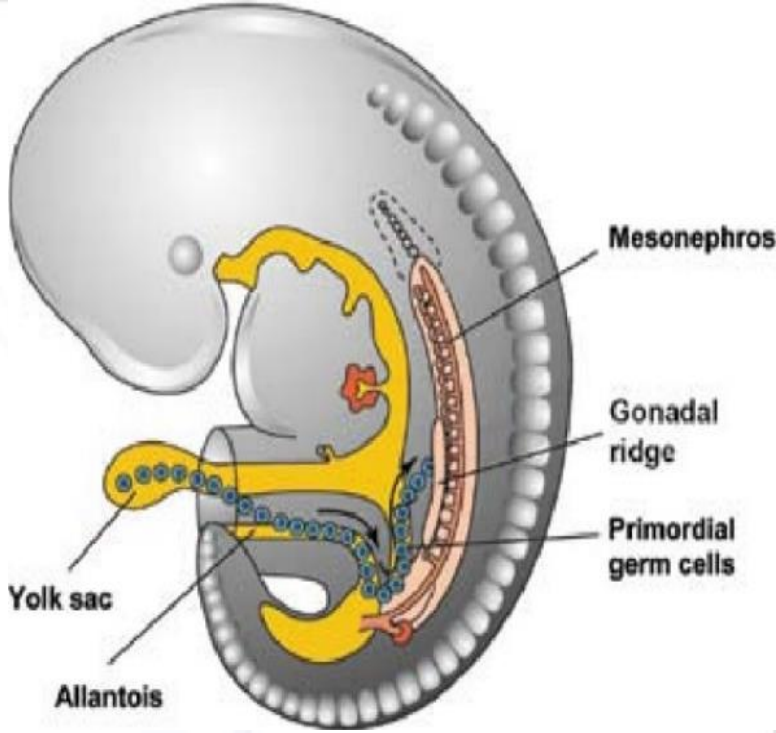
The Indifferent Embryo

Gonads appear initially as a pair of longitudinal ridges known as the genital or gonadal ridges. They are formed by proliferation of the **coelomic epithelium** and a condensation of underlying **mesoderm** forming the primitive sex cords.





Development of the gonads
The Indifferent Embryo



Primordial germ cells (PGCs) :

- They originate from the wall of the yolk sac and migrate to invade the genital ridges and sex cords in the sixth week.

- The primordial germ cells have an **inductive influence** on development of the gonads

